

CHI WAI CHAN

Hong Kong

☎ +852-9442-5274 ✉ chanchiwairay@gmail.com 🌐 chi-wai-chan-ba5870137 📺 chanchiwai-ray

Summary

Platform Software Engineer with 4 years of experiences, specializing in open source distributed systems, cloud-native orchestration, security, and observability. Proven track record at engineering Juju charmed operators (Infrastructure-as-Code) and native telemetry in Golang and Python to help manage highly available OpenStack and Kubernetes clusters at scale. Bridge the gap between low-level infrastructure development and enterprise reliability requirements.

Experience

Canonical

May 2022 – Present

Graduate DevOps Engineer, Associate DevOps Engineer, Software Engineer I

Hong Kong, China

- **Infrastructure Automation:** Engineered and maintained a Python-based automation framework (charmed-openstack-upgrader) to orchestrate zero-downtime cluster upgrades across distributed OpenStack environments, heavily reducing manual operational overhead for enterprise clients.
- **Infrastructure Orchestration:** Engineered Juju charmed operators (using Python) to deploy, manage, and scale complex cloud infrastructure across Kubernetes and OpenStack environments.
- **Infrastructure Maintenance:** Automated infrastructure maintenance workflows for OpenStack, Kubernetes, and MicroCeph environments. Reduced the need for tedious and error prone manual maintenance workflow.
- **Infrastructure Observability:** Observability instrumentation for OpenStack environments by building custom Prometheus exporters (e.g. Hardware Observer Operator and OpenStack Exporter Operator). Enabled real-time telemetry, alerting, and dashboarding for enterprise hardware and cloud control planes.
- **Security:** Architected secure and immutable container packages (Snaps, Rocks) for distributed applications, ensuring secure, cross-platform software delivery.
- **Product Engineering Feedback Loop:** Acted as the technical bridge for the Managed Solution team, analyzing real-world enterprise infrastructure bottlenecks and directly contributing fixes and enhancements back into upstream core products (Sunbeam, Kafka, MicroCeph).

Research Center for the Early Universe, University of Tokyo

Aug 2021 – Apr 2022

Project Researcher

Tokyo, Japan

- Improved GstLAL pipeline's offline dashboard to support on-demand loading of images.
- Developed a static group website and prepared TensorFlow tutorial for new members in the group.
- Helped re-analyze gravitational wave with the latest GstLAL pipeline at the time, and ensured the quality of the pipeline.

HENNGE

Jul 2021 – Aug 2021

Software Engineer Intern

Tokyo, Japan

- Contributed a social media web application using Python (Flask framework).
- Implemented authentication and login system with Google OAuth 2.0 and Redis.
- Implemented text and image storage system using Redis and MinIO.
- Employed micro-services architecture and deployed the web application using Docker, AWS, and Terraform.

Education

The University of Tokyo

Sep 2018 – Sep 2020

Master of Science in Physics

Tokyo, Japan

- Overall grade A in course work performance
- Teaching Assistant in Gravitational Wave Physics
- Submitted a technical paper to arXiv (arXiv:2009.03025)

- Dean's List 2015 - 2017
- HKUST Robust team (Smart Car team 2016 - 2017)
- Graduated with First Honor Class (GPA 3.7/4.3)

Technical Skills

- **Languages:** Python, Golang (Go), Bash, TypeScript
- **Cloud Native:** Kubernetes, LXD, Docker, Rocks
- **Infrastructure:** OpenStack, Ceph (MicroCeph), Kafka, Juju, Terraform, Terragrunt, Snaps
- **Observability (SRE):** Prometheus, Grafana, Loki, OpenTelemetry, custom exporters
- **CI/CD and Operations:** GitHub Actions, Linux/Ubuntu, Git

Certifications

AWS Certified Cloud Practitioner Amazon Web Services Training and Certification	December 2025 - December 2028 LF-ehjghdyant
Certified Kubernetes Administrator The Linux Foundation	October 2025 - October 2027 LF-effnrqhxfp
Linux Foundation Certified Systems Administrator The Linux Foundation	June 2025 - June 2027 LF-ehjghdyant
Certified Kubernetes Application Developer The Linux Foundation	Nov 2024 - Nov 2026 LF-8023d8bwms

Achievements

Masanori Nagashima Scholarship	2019
Dr Sampson Sun Scholarship Fund	2018
Third place winner at 12th NXP Cup Intelligent Car Racing Competition (South China Region)	2017

Languages

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|-------------|-----------|------------|-----------------------|
| • Cantonese | • English | • Mandarin | • Japanese (N2 level) |
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